FINAL-YEAR RESEARCH PROJECT · SUMMARY

Detecting Fake Kithul Treacle with Low-Cost AI

A multi-method system using phone camera vision, a low-cost sensor device and thermal imaging to detect and measure sugar adulteration in Sri Lankan kithul treacle.

1. What is the problem?

Kithul treacle (kithul pani) is a traditional Sri Lankan sweetener made from the sap of the kithul palm. In December 2025, UNESCO recognised kithul tapping as part of the world's cultural heritage.

But there is a big problem: **many sellers mix cheap sugar syrup into kithul treacle and sell it as pure.**

- Buyers pay a high price but often get a fake or mixed product.
- Honest tappers lose income because fake treacle is cheaper to make.
- Sri Lanka's exports suffer because foreign buyers lose trust.
- People who choose kithul as a "healthier" sweetener may be eating mostly sugar.

Today, the only reliable way to check treacle is a chemical lab test. These tests are slow and expensive, and ordinary buyers, shops and small sellers can't do them. **There is no quick, cheap and easy way to check whether kithul treacle is pure.**

2. Is there existing research?

Yes, but none of it solves this problem fully.

Research on kithul treacle

- **Seneviratne and Dissanayake (2016)** surveyed 304 tappers in 101 villages. They found sugar is mainly added when the sap is poor or watery, especially in rainy weather.
- **Somasiri (2015, University of Colombo)** studied the chemistry of kithul using lab machines (HPLC, GC-MS) and made a chemical test kit to detect cane sugar.
- **Wijesinghe and Rathnayake (2021, Wayamba University)** mixed sugar syrup into pure treacle (5% to 25%). They showed that pure and mixed treacle differ in properties like sugar level (Brix), pH and electrical conductivity. They said no proper testing method had been established yet.

Research on honey, which has the same problem

- A Thai study used a cheap light sensor with machine learning and detected sugar syrup in honey with 91-98% accuracy.
- A Moroccan study used a thermal camera and deep learning (CNN) and detected mixed honey with about 99% accuracy.

3. What is the research gap?

- Kithul research so far uses **only chemical lab methods**, which are slow, costly and need experts.
- Cheap AI-based methods **already work for honey**, but **nobody has tried them on kithul treacle**.
- **No dataset** of kithul treacle images, sensor readings or videos exists.
- Nobody has compared which low-cost method works best for kithul, or how cheap a reliable test can be.

> **In one sentence:** There is no fast, low-cost, AI-based method to detect and measure sugar adulteration in kithul treacle, and no dataset to build one.

4. How are we going to do this?

We will build a system that checks kithul treacle using **three different low-cost methods**, then combine them.

1. **Collect samples.** Get pure kithul treacle from trusted tappers and confirm it is pure with a lab test.
2. **Make mixed samples.** Mix the pure treacle with sugar syrup at known amounts: 0%, 5%, 10%, 15%, 20%, 30% and 50%. Optionally, also try other cheap syrups, such as jaggery syrup or coconut treacle.
3. **Measure every sample in three ways:** with a phone camera (images and video), with a small sensor device, and with a thermal camera.
4. **Train AI models.** Each method gets its own model to predict whether the treacle is pure or mixed, and how much sugar was added (%).
5. **Combine the three methods** into one final model and check whether it is more accurate than any single method.
6. **Test on real products.** Buy treacle from shops and supermarkets, test it with our system, then confirm with a lab test.
7. **Compare cost and accuracy.** Find the cheapest method that is good enough for each type of user: consumers, shops and exporters.

Expected result: a cost-vs-accuracy comparison

Level	Who uses it	Method
Cheapest	Consumers	Phone camera only
Middle	Collectors, shops	Handheld sensor device
Best	Exporters, certifiers	All three methods combined

5. Where do we find the dataset?

No ready-made dataset exists, so we will create our own. This is also one of our research contributions.

Where to get the samples

- **Pure treacle:** directly from kithul tappers in kithul-growing areas, or through the **Kithul Development Board**, which works with tappers.
- **Market samples:** from shops, supermarkets, roadside sellers and online sellers.
- **Sugar syrup:** we make it ourselves from ordinary table sugar, so we know the exact mixing amount.

Who confirms the samples are pure

- A food science or chemistry lab, such as a university chemistry department or the **Industrial Technology Institute (ITI)**.
- It's worth contacting the researchers from the Wayamba University and University of Colombo studies, since they have already done similar lab work on kithul.

About how much data

Roughly 10-15 pure samples × 7 mixing levels × 2-3 types of syrup × 3 repeats. That gives about **400–1,300 measurements per method**.

Help for small data

Use pre-trained AI models (transfer learning) and data augmentation, so the models work even with a limited dataset.

6. How do we divide the modules?

Each person builds a different method to detect fake treacle. All three test the same samples.

Member	Module	Main output
Member 1	Phone camera vision	Phone app test for consumers
Member 2	Low-cost sensor device	Pocket tester for shops and collectors
Member 3	Thermal camera + combined model	Most accurate system for exporters

Member 1: Phone camera module

Question: Can an ordinary smartphone detect fake treacle?

- Build a small light box so photos are taken in the same lighting.
- Take photos to study colour and clarity.

- Record videos of treacle flowing or dripping, because added sugar changes how thick it is.
- Use computer vision to measure the flow: tracking the drop and its speed and spread.
- Train a CNN model for images and a video model (CNN + LSTM) for flow.
- **Output:** a phone app test that anyone can use.

Member 2: Sensor device module

Question: Can a cheap handheld device detect fake treacle as well as a lab?

- Build a small device with an ESP32 board, a multispectral light sensor (such as AS7265x) and an electrical conductivity sensor.
- Collect light readings and conductivity readings from each sample.
- Train models: PLS and SVM, compared with a 1D-CNN.
- **Output:** a low-cost pocket tester for shops and collectors.

Member 3: Thermal camera and combined model module

Question: Can heat patterns detect fake treacle, and does combining all methods give the best result?

- Use a thermal camera (a low-cost MLX90640 sensor, or a borrowed FLIR camera).
- Record how each sample heats up or cools down, since pure and mixed treacle behave differently.
- Train a deep learning model on the thermal videos (ConvLSTM or 3D CNN).
- Build the **combined (fusion) model** that uses results from all three methods.
- Explain the model's decisions using Grad-CAM and SHAP.
- **Output:** the most accurate combined system for exporters and certifiers.

References

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Note: open each reference and check the details before final submission. Some papers were reviewed from their abstracts only.